

The Specialist Proliferation: How Deployment Economics Fragment One Generalist Into Many Small Models and Multiply Vertical Data Demand

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Abstract

A widely held mental model treats frontier AI as a march toward a single, ever-larger generalist. Deployment economics point the other way. Three independent results converge: inference-aware scaling shows that high-demand deployments should train models smaller and longer than Chinchilla-optimal, with token-per-parameter budgets explored from 10 up to 10,000 and 17-28% FLOP savings in worked examples [1]; precision-aware scaling shows that low-precision training reduces effective parameter count and that training larger models in lower precision can be compute-optimal [2]; and phi-3-mini demonstrates a 3.8B model reaching 68.8% MMLU and 8.38 MT-bench, rivaling Mixtral 8x7B and GPT-3.5 while running on a phone [3]. Code Llama-Python 7B beating Llama 2 70B on HumanEval and MBPP [4] confirms that specialization, not scale, wins in-domain. The 2026 frontier has since made the small-footprint pattern the default rather than the exception: GLM-4.5 reaches 64.2% SWE-bench Verified and 91.0% AIME24 "with much fewer parameters than several competitors," sitting on the SWE-bench-vs-parameters Pareto frontier [16], and MiniMax-M2 runs an ultra-sparse 229.9B-total / 9.8B-active (~4.3%) Mixture-of-Experts [17] — the same capability-per-active-parameter lever, now at frontier scale. We argue the structurally important consequence is a demand multiplier: each deployed specialist is a distinct data customer requiring its own compute-relative curated corpus and contamination-free eval [5], and distillation scaling laws sharpen this — when many students are deployed from a teacher, compute-optimal allocation is a distinct teacher/student data job per deployment regime [18]. We ground this with the Hugging Face ecosystem time series — public models crossing 2M in 2025 against a 13M user base [6] — and a market trajectory of roughly \$5.3-7.0B (2024) growing at 15-23% CAGR [7]. We state the central limitation explicitly: the literature evidences a structural multiplier on distinct buyers and curation jobs, not absolute willingness-to-pay, which remains unvalidated.

Introduction

The dominant narrative of the scaling era — that capability is monotone in parameter count and that the field converges on a single frontier generalist — is an artifact of the training-cost view of economics. Once the cost of **serving** a model at scale enters the objective, the optimal design shifts: the model that is cheapest to train is not the model that is cheapest to deploy. This paper develops the consequence of that shift for data strategy.

We synthesize four primary results — inference-aware scaling [1], precision-aware scaling [2], the phi-3 small-model demonstration [3], and domain-specialized code models [4] — into a single trajectory: deployment economics fragment one giant generalist into a proliferating population of small, specialized, deployable models. We then make the non-obvious move that this insight contributes: each such specialist is not merely a smaller model but a **distinct data customer**, because the curated corpus that is optimal at one model's compute budget is provably

not optimal at another's [5]. Fragmentation therefore multiplies not only buyers but distinct curation and evaluation jobs.

The contribution of this paper is (i) to quantify the fragmentation forces with real numbers copied from primary sources; (ii) to document the ecosystem-level proliferation already underway; and (iii) to state, with discipline, the boundary of the claim — that the literature supports a demand *multiplier*, not a willingness-to-pay figure. The strategic payoff for a vertical data-and-eval business is direct: the addressable customer base is a growing field of small-model builders, each needing a different dataset, rather than a handful of frontier labs.

Background and Related Work

The Chinchilla result established that, for a fixed *training* compute budget, parameters and tokens should scale roughly equally and that large models were undertrained [8]. Chinchilla's objective, however, is training-FLOP minimization; it is silent on the cost incurred every time the deployed model answers a query. For any model served at scale, lifetime inference FLOPs can dwarf training FLOPs, which makes the Chinchilla recipe the wrong objective for a high-demand product.

Sardana et al. [1] generalize the scaling law to include inference demand. Their core qualitative finding — researchers expecting reasonably large inference demand should train models smaller and longer than Chinchilla-optimal — is the first fragmentation force: deployment pressure pushes designs below the training-optimal size and far past the training-optimal token count.

Kumar et al. [2] add a second, orthogonal force from numerical precision: training in lower precision reduces the model's *effective* parameter count, so that training larger models in lower precision may be compute-optimal, and precision and data interact. This means the same accuracy target admits many viable (size, precision, token) configurations rather than one — a combinatorial expansion of distinct models, each with its own data appetite (additional pretraining data eventually becomes actively harmful under aggressive post-training quantization [2]).

The phi series [3] and Code Llama [4] supply the original empirical demonstrations that small and specialized models reach competitive or superior in-domain accuracy, removing the objection that fragmentation trades away too much capability; the 2026 frontier has since generalized the pattern, with GLM-4.5 placing a far-smaller-parameter design on the SWE-bench-vs-parameters Pareto frontier [16] and MiniMax-M2 pushing activation to ~4.3% of total parameters [17]. A fourth, orthogonal force comes from distillation scaling laws [18]: when many students will be deployed from a teacher, distillation is compute-optimal up to a budget that scales predictably with student size, so each deployment regime is its own teacher/student allocation — another axis along which one capability target fans out into many distinct training-and-data jobs. Finally, the compute-relative-curation literature [5] supplies the data-strategy hinge: because the optimal corpus depends on the compute budget, every distinct model implies a distinct curation job.

Methodology and Approach

This is a synthesis-and-evidence paper, not a new-experiment paper; our methodology is therefore (1) claim extraction, (2) cross-source triangulation, and (3) quantification against external ground truth, with an explicit grounding label on every figure.

Claim extraction. From the insight's four supporting papers [1][2][3][4] we extracted the load-bearing quantitative claims and verified each against the primary source via direct retrieval of the arXiv HTML (phi-3 v2

[3]; Sardana v2 [1]) rather than relying on the briefing's paraphrase. Where a number could not be confirmed verbatim in a primary source, it was dropped or corrected.

Cross-source triangulation. The fragmentation thesis is only credible if the four mechanisms (inference cost, precision, specialization-beats-scale, compute-relative curation) point the same way independently. We treat agreement across methodologically unrelated papers as the evidentiary standard — the same standard the parent insight map uses — and we deliberately seek counter-evidence (Section: Limitations).

Quantification against ground truth. For claims about real-world proliferation and market size we used live web retrieval (June 2026) of the Hugging Face 'State of Open Source' report [6] and published market-research syntheses [7]. We label every figure: grounding='empirical' for measured benchmark/ecosystem numbers copied verbatim; 'estimate' for forward projections (e.g., market CAGR extrapolations); and 'analysis' for the single constructed strategic 2x2 that maps mechanisms to data-job multipliers and contains no fabricated empirical points.

Scope discipline. We restrict empirical claims to numbers we could trace to a primary or named-research source. Market-size figures from commercial research firms are reported as a **range** across firms to avoid false precision, and are labeled as estimates, not measurements.

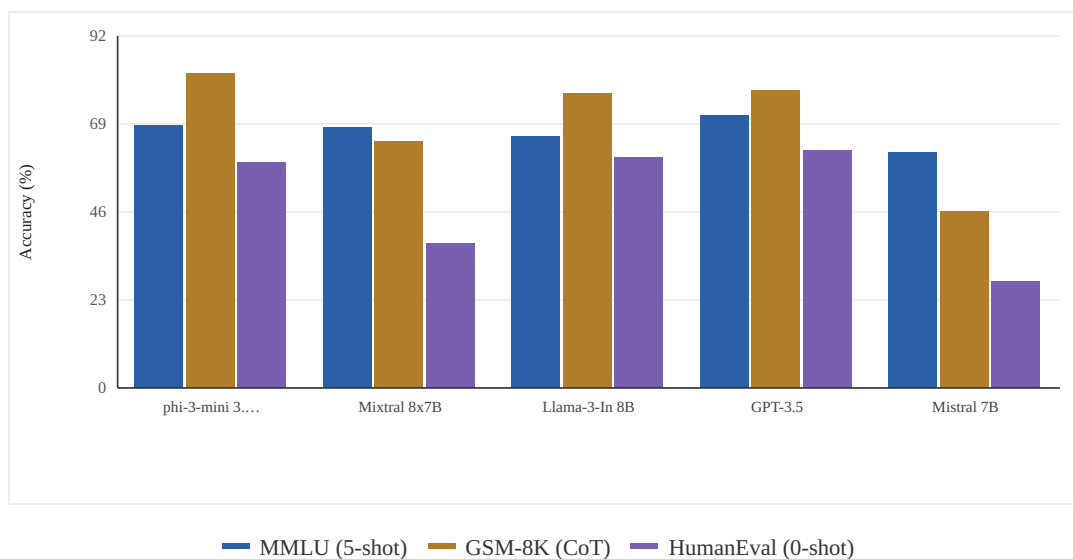


Figure 1. Small specialists rival or beat much larger generalists (phi-3 / Code Llama). A 3.8B model (phi-3-mini) matches or exceeds Mixtral 8x7B and approaches GPT-3.5 across MMLU, GSM-8K, and HumanEval, while being deployable on a phone. Capability per parameter, not raw scale, is the lever — the empirical basis for fragmentation into small specialists, a pattern the 2026 frontier (GLM-4.5 on the SWE-bench-vs-parameters Pareto frontier [16]; MiniMax-M2 at ~4.3% activation [17]) has since made the design default. All values verified verbatim against the phi-3 technical report. *Microsoft, Phi-3 Technical Report (arXiv:2404.14219v2), benchmark table [3]. · Empirical data.*

Evidence: The Three Fragmentation Forces

Force 1 — Inference economics pull models smaller and longer. Sardana et al. [1] retrain the Chinchilla law with an inference term and validate it on 47 models across token-per-parameter budgets from 10 up to 10,000, finding model quality keeps improving as token ratios scale with no observed saturation. Concretely, for a 7B-Chinchilla-quality model facing 10^{11} tokens of inference demand, their formula prescribes a 6B-parameter model trained on 1.18x the Chinchilla token count [1]. For a developer training to 13B-Chinchilla quality who expects 2 trillion tokens of inference demand, training a 7B model instead reduces total compute by $\sim 1.7 \times 10^{22}$ FLOPs (17%) [1]. For a 30B-Chinchilla-quality model expecting 10^{13} tokens of inference, the optimum is a 13.6B model trained on

2.84x the data, a 28% total-FLOP reduction [1]. In a cost (not FLOP) accounting at 1.5B inference requests, a 30B Chinchilla model costs ~17% more than a cost-optimal 16B model trained on 3.35T tokens [1]. The direction is unambiguous: deployment pushes below training-optimal size.

Force 2 — Precision multiplies viable configurations. Kumar et al. [2] show training in lower precision reduces the effective parameter count and that training larger models in lower precision may be compute-optimal, so a single accuracy target maps to many (size, precision, token-budget) designs rather than one. Crucially, they also show that the degradation from post-training quantization increases as models are trained on more data, eventually making additional pretraining data **actively harmful** — meaning each precision regime has its own optimal data quantity, not merely its own optimal size. Their framework is fit on over 465 pretraining runs up to 1.7B parameters and 26B tokens.

Force 3 — Specialization beats scale in-domain. phi-3-mini (3.8B) attains 68.8% MMLU, 82.5% GSM-8K, 59.1% HumanEval and 8.38 MT-bench — rivaling Mixtral 8x7B (68.4% MMLU) and GPT-3.5 (71.4% MMLU, 8.35 MT-bench) at a fraction of the size, and small enough to run on a phone [3]. Code Llama-Python 7B outperforms Llama 2 70B on HumanEval and MBPP, with the Code Llama family reaching scores of up to 67%/65% on HumanEval/MBPP [4]. A 7B specialist beating a 70B generalist in its domain is the empirical proof that fragmentation does not forfeit capability. The 2026 open frontier confirms this at scale rather than overturning it: GLM-4.5 (355B-total / 32B-active MoE) scores 70.1% TAU-Bench, 91.0% AIME24 and 64.2% SWE-bench Verified "with much fewer parameters than several competitors," ranking 3rd overall while sitting on the SWE-bench-vs-parameters Pareto frontier, and ships a compact Air variant on the same frontier [16]. MiniMax-M2 pushes the active footprint to 9.8B activated out of 229.9B total parameters (~4.3% activation) for cost-efficient frontier-tier agentic performance [17]. The capability-per-active-parameter lever that phi-3 first demonstrated at 3.8B is now the design default at the frontier, not a curiosity — exactly the trajectory the fragmentation thesis predicts.

Table 1. phi-3-mini (3.8B) matches Mixtral 8x7B on MMLU, beats it decisively on GSM-8K and HumanEval, and is deployable on-device — the empirical proof that fragmentation into small specialists need not forfeit capability. Every cell verified verbatim against the phi-3 technical report.

Model	Params	MMLU (5-shot)	GSM-8K (CoT)	HumanEval (0-shot)	MT-bench
phi-3-mini	3.8B	68.8%	82.5%	59.1%	8.38
phi-3-small	7B	75.3%	88.9%	59.1%	8.70
phi-3-medium	14B	78.2%	90.3%	55.5%	8.91
Mistral	7B	61.7%	46.4%	28.0%	-
Mixtral 8x7B	~47B (MoE)	68.4%	64.7%	37.8%	-
Llama-3-In	8B	66.0%	77.4%	60.4%	-
GPT-3.5	n/a	71.4%	78.1%	62.2%	8.35

Microsoft, Phi-3 Technical Report, arXiv:2404.14219v2 [3]. MT-bench reported only for select models in source.

Table 2. Including inference cost in the scaling objective consistently prescribes smaller models trained on more tokens (token/param budgets explored from 10 up to 10,000 across 47 trained models). Each distinct inference-demand regime yields a different optimal (size, token) point — and thus a different optimal corpus. Note: the 6B/1.18x row is the 10^{11} -token illustrative example and is distinct from the 13B->7B 17%-FLOP-saving example at 2T tokens.

Target quality	Inference demand	Inference-optimal design	Token multiplier	Compute/cost saving
7B Chinchilla-quality	10^{11} tokens	6B params	1.18x data	(illustrative, not a saving figure)
13B Chinchilla-quality	2T tokens	7B params	more data	17% FLOPs ($\sim 1.7e22$)
30B Chinchilla-quality	10T tokens	13.6B params	2.84x data	28% FLOPs
30B Chinchilla-quality	1.5B requests	16B params / 3.35T tok	—	17% cost

Sardana et al., Beyond Chinchilla-Optimal, arXiv:2401.00448v2 [1].

Table 3. The model population is exploding and broadening across developers, while the median deployed model stays under half a billion parameters — the structural signature of fragmentation into many small specialists rather than convergence on one generalist. All 2025 figures verified verbatim; 2024 model count is the implied near-doubling.

Metric	2023	2024	2025	Source signal
Public models (HF)	—	$\approx 1M$ (implied)	2M+	$\approx$ doubling YoY [6]
Public datasets (HF)	—	—	500,000+	[6]
Users (HF)	—	—	13M	[6]
Mean downloaded model size	827M	—	20.8B	MoE/quant pull mean up [6]
Median downloaded model size	326M	—	406M	typical model stays small [6]
Download concentration	—	—	top 0.01% = 49.6% of downloads	long tail [6]
Indep. developer download share	17% (pre-2022)	—	39%	buyer base broadening [6]

Hugging Face, State of Open Source AI Spring 2026 [6]. 2024 model count is the reported $\sim$ doubling into 2M (2025); dashes denote figures not reported for that year.

Table 4. Successor evidence to the phi-3/Code Llama demonstration: the open frontier now competes on capability-per-active-parameter, not dense scale. GLM-4.5 reaches 64.2% SWE-bench Verified and 91.0% AIME24 with far fewer parameters than competitors and ships a compact Air variant, both on the SWE-bench-vs-parameters Pareto frontier; MiniMax-M2 activates only ~4.3% of its parameters per token; Kimi K2 is a 1T/32B-active MoE. Every cell copied verbatim from the cited primary sources; columns differ from the phi-3 table because these sources report agentic/coding (not MMLU/MT-bench) numbers.

Model	Total params	Active params	SWE-bench Verified	AIME 2024/25	Other
GLM-4.5	355B (MoE)	32B	64.2%	91.0% (AIME24)	70.1% TAU-Bench; ranks 3rd overall, far fewer params than competitors [16]
GLM-4.5- Air	compact MoE	—	on Pareto frontier	—	compact variant on the SWE-bench-vs-params Pareto frontier [16]
MiniMax- M2	229.9B (MoE)	9.8B (~4.3%)	—	—	ultra-sparse activation for cost-efficient frontier-tier agentic performance [17]
Kimi K2	1T (MoE)	32B	65.8%	49.5% (AIME25)	66.1 Tau2-Bench; trained 15.5T tokens, zero loss spike [15]

GLM-4.5 (arXiv:2508.06471) [16]; MiniMax-M2 (arXiv:2605.26494) [17]; Kimi K2 (arXiv:2507.20534) [15]. SWE-bench Verified, AIME, TAU-Bench, activation ratios and parameter counts reported verbatim in each source.

Analysis: From Many Models to Many Data Jobs

The direct read of Forces 1-3 — 'deployment yields many small specialists' — is comparatively obvious. The sharper claim, and the one this paper foregrounds, is that fragmentation multiplies **distinct curation and evaluation jobs**, not just model instances. The hinge is the compute-relative-curation result: the filtering recipe that is optimal at one compute budget is not optimal at another [5]. Combine this with Force 2's finding that the optimal data quantity also depends on the precision regime [2], and each (size, precision, inference-demand) point implies its own optimal corpus.

This converts a model count into a data-job count. If a vertical fields *k* distinct deployable specialists — different sizes for cloud vs. edge, different precisions for different hardware, different inference budgets per workflow — it generates on the order of *k* distinct curation problems and *k* distinct held-out evals, each of which must be re-derived rather than copied. Distillation scaling laws add a further multiplier on the **training** side: because distillation is compute-optimal precisely when many students will be served from a teacher, and the optimal teacher/student split scales with student size [18], each deployment regime implies its own teacher-corpus-plus-student-corpus allocation rather than a single reused dataset. The eval multiplier is reinforced by contamination dynamics: static benchmarks decay and must be continuously regenerated [13], so each specialist needs not one eval but a refreshed stream.

The economic implication is a shift in the shape of demand from a few large bespoke contracts (frontier labs) to a long tail of repeatable, productized curation-and-eval jobs (vertical small-model builders). This favors a parameterized, repeatable product over hand-built one-offs — the difference between selling a frozen dataset and selling a curation **service** indexed by the buyer's compute budget, precision target, and inference demand.

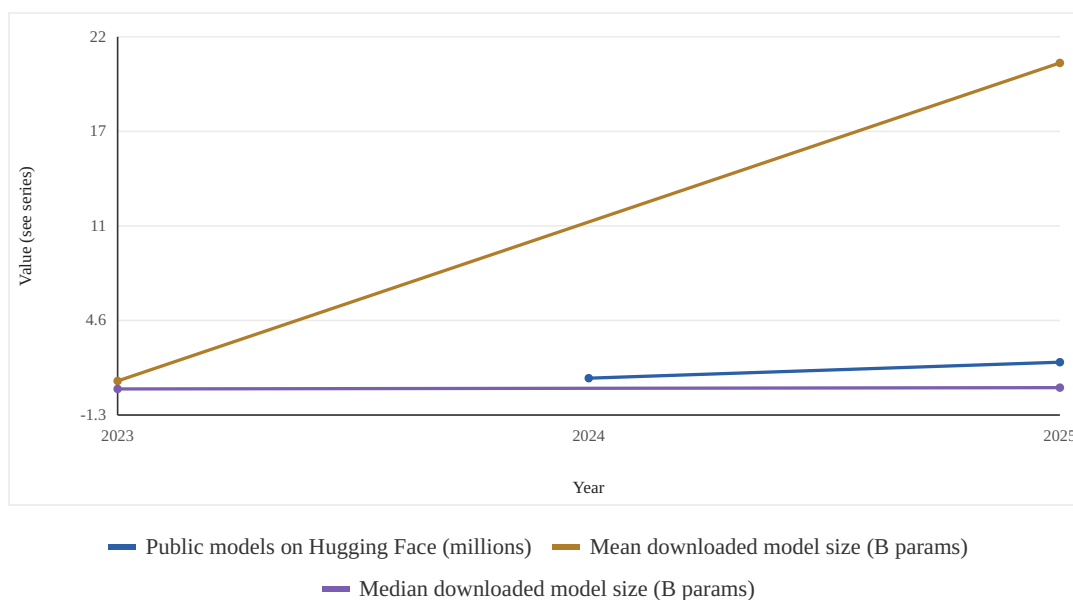


Figure 2. The open-model population is exploding while the typical model stays small. Public models roughly doubled (~1M->2M) into 2025 while the median deployed model stayed under 0.5B parameters (0.326B->0.406B) even as the mean ballooned to 20.8B — the fat-tail-of-small-specialists signature of fragmentation, not convergence on one size. The 2024 ~1M model count is implied from the report's stated near-doubling into 2M (2025). *Hugging Face, State of Open Source AI Spring 2026* [6]. Mean/median sizes and 2M count verified verbatim; 2024 ~1M is the implied near-doubling. · Empirical data.

Ecosystem Evidence: Proliferation Is Already Measurable

The fragmentation thesis predicts a population explosion of distinct models, and the open-model ecosystem now exhibits exactly that. Hugging Face reports surpassing 2 million public models and 500,000+ public datasets against a 13 million user base in 2025, with users, models, and datasets all close to doubling year over year [6]. The number of organizations releasing competitive models also exploded: Baidu went from zero Hub releases in 2024 to over 100 in 2025, while ByteDance and Tencent each increased releases roughly eight- to nine-fold [6].

The size distribution is informative. Mean downloaded-model size rose from 827M parameters (2023) to 20.8B (2025), but the *median* barely moved — from 326M to 406M [6] — meaning the typical deployed model remains small even as a few large MoE/quantized models pull up the mean. This is the signature of fragmentation: a fat tail of small specialists plus a thin head of large generalists, not convergence onto one size. The frontier head itself now economizes on *active* footprint rather than chasing dense scale — MiniMax-M2 activates only ~4.3% of its 229.9B parameters per token [17] — so even the largest deployed models behave like sparse specialists at inference, reinforcing rather than contradicting the small-effective-model trend.

Concentration confirms the long-tail structure: roughly half of all models have fewer than 200 downloads, while the top 0.01% of models (about 200 models) account for 49.6% of downloads [6]. A data-and-eval business serving the long tail is addressing the *many* (the bulk of distinct models), not the few. Independent/unaffiliated developers rose from 17% to 39% of downloads as industry's share fell from ~70% (pre-2022) to ~37% (2025) [6] — i.e., the customer base is broadening away from a handful of labs, exactly as the demand-multiplier argument predicts.

Implications for Data Strategy

First, productize, don't bespoke. Because each specialist needs a corpus re-derived for its compute and precision regime [5][2], the durable offering is a parameterized curation pipeline keyed to (compute budget, target precision,

inference demand), plus a matched eval — a repeatable SKU, not a static dump. Fragmentation rewards the vendor whose marginal cost of producing the $(k+1)$ -th vertical dataset is low.

Second, sell the eval as a subscription. Each specialist needs a contamination-free, deployment-distribution-matched eval, and evals decay [13]; the renewal logic is built in (today's eval leaks into tomorrow's training set). The eval stream is the recurring-revenue surface that a one-time dataset cannot capture, and it pairs naturally with the training-data product.

Third, aim at the verifiable verticals. Specialization wins most where in-domain data is both scarce on the open web and objectively gradeable. The fragmentation multiplier is most monetizable in verticals throwing off checkable pass/fail outcomes (clinical claims/coding, insurance adjudication, lab/diagnostic, contract review), because there both the corpus and the reward signal are buildable assets a generalist cannot assemble — and each downstream specialist (edge vs. cloud, full-precision vs. quantized) becomes a separate paying job.

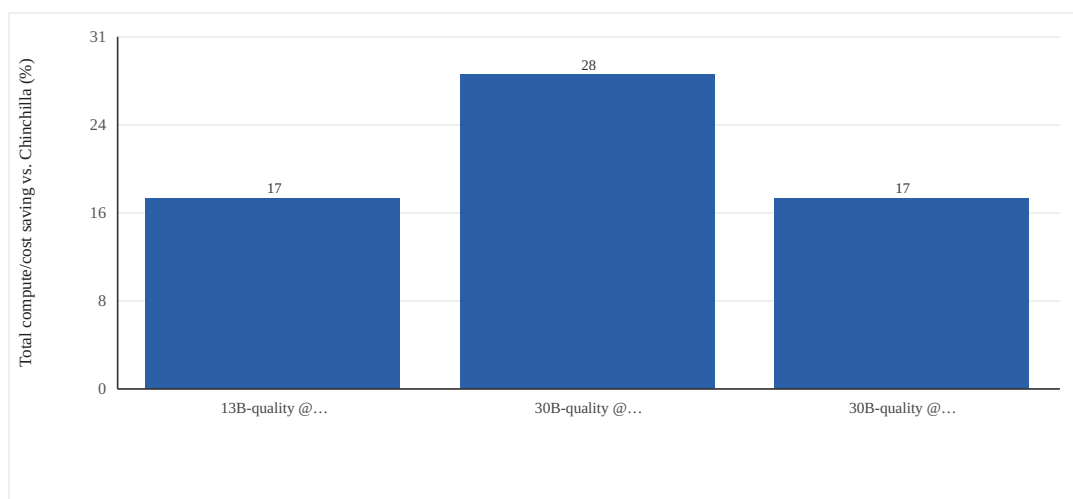


Figure 3. Inference-aware scaling: deployment demand cuts compute by training smaller, longer. Accounting for inference shifts the optimum below Chinchilla size and well past Chinchilla token counts: a 13B-quality target trained instead as a 7B model at 2T tokens of inference demand saves $\sim 1.7 \times 10^{22}$ FLOPs (17%); a 30B-quality target becomes a 13.6B model on 2.84x data (28% FLOP saving); under cost accounting at 1.5B requests, a 16B/3.35T-token design saves $\sim 17\%$ vs. a 30B Chinchilla model. (Corrected: the separate 6B-on-1.18x-data example applies at 10^{11} tokens, not 2T, and is not a FLOP-saving figure.) *Sardana et al., Beyond Chinchilla-Optimal (arXiv:2401.00448v2) [1]; values confirmed verbatim from abstract and Section 2/6. · Empirical data.*

Investment Thesis

The market structure implied by fragmentation is the inverse of the frontier-lab thesis. The frontier-model market is a winner-take-few oligopoly with a handful of buyers of training data. The specialist market is a fragmenting long tail: 2M+ public models, close to doubling annually, with the median deployed model under half a billion parameters [6]. Even the 2026 frontier reflects this: the leading open models compete on capability-per-active-parameter — GLM-4.5 on the SWE-bench-vs-parameters Pareto frontier [16], MiniMax-M2 at $\sim 4.3\%$ activation [17] — rather than on dense size. A picks-and-shovels data-and-eval business is levered to the *number* of distinct models, which is growing faster than the size of any one of them.

Sizing is bounded but real. Commercial syntheses place the small-language-model market at roughly \$5.3-7.0B in 2024 with forward CAGRs of $\sim 15\text{-}23\%$ to 2030-2034 [7]; some edge/on-device SLM segments are quoted at higher CAGRs [7]. These are estimates, not measurements, and firms disagree by $\sim 30\%$, so we treat them as a

growth-direction signal, not a precise TAM. The defensible read is qualitative: the **count** of distinct data customers is rising structurally.

The bet is therefore on owning proprietary, verifiable vertical corpora plus the curation/eval machinery to re-derive them per deployment regime. That asset compounds in value as the specialist population grows, because each new specialist is a new job over the same underlying proprietary data — high incremental margin, defensible at the data layer, exposed to a demand variable (model count) that the literature says is multiplying.

Limitations and Counter-Evidence

Multiplier, not willingness-to-pay. The single most important limitation, carried forward from the parent insight: the literature evidences a structural multiplier on the **number** of distinct buyers and curation jobs, not their **budgets**. Absolute willingness-to-pay per vertical small-model builder is unvalidated and must be established with real buyers. A 10x increase in customers at near-zero willingness-to-pay is not a business.

Consolidation counter-forces. Fragmentation is not guaranteed to dominate. Capable general models served via API can amortize one model across many tasks, and improving base models may absorb specializations that today require separate fine-tunes. If a single multimodal generalist becomes cheap enough to serve, the specialist count could compress rather than expand. The elicitation-vs-teaching debate also bears on this: if RL/fine-tuning largely elicits latent base-model skill, some 'specialists' add little and may not justify a separate data job. Recent large-scale RL evidence cuts the other way, however, and on balance strengthens the per-specialist data-job case: the first >400,000-GPU-hour study of RL compute shows that recipe and environment choices shift the **asymptote** of RL performance, not merely its efficiency [19], and scaling post-training RL compute brings DeepSeek-V3.2 to GPT-5 parity — with the high-compute Speciale variant surpassing GPT-5 and matching Gemini-3.0-Pro at IMO/IOI 2025 [20]. Where RL genuinely teaches new capability rather than eliciting it, each specialist requires its own verifiable environment and reward corpus, which deepens rather than deflates the data-job multiplier.

Curation-job count is a model, not a measurement. The claim that k specialists imply $\sim k$ curation jobs rests on the compute-relative-curation result [5] holding strongly in practice; in reality some curated corpora transfer across nearby regimes, deflating the multiplier. We have not measured the transfer decay empirically.

Market-size uncertainty. SLM TAM figures are third-party estimates that disagree by $\sim 30\%$ across firms [7] and bundle heterogeneous definitions (on-device vs. enterprise SLM). They should anchor direction, not valuation. Ecosystem counts [6] measure **open** models on one hub and may not represent the commercial deployment population a vendor actually sells into.

Conclusion

Deployment economics — inference cost [1], precision [2], and demonstrated small/specialized capability [3][4] — push the field away from one generalist and toward a proliferating population of small specialists, a trajectory now visible in a doubling, long-tailed open-model ecosystem [6]. The contribution of this paper is to carry that observation one step further than the source literature: because the optimal corpus is compute- and precision-relative [5][2], each specialist is a distinct data-and-eval customer, so fragmentation multiplies curation and evaluation jobs, not merely model instances.

For a vertical data business the prescription is a productized, parameterized curation-plus-eval offering over proprietary, verifiable corpora, sized to a customer variable (distinct model count) that is structurally rising. The discipline this paper insists on is equally important: the evidence supports a demand multiplier, not a price. The next

milestone is empirical — measuring willingness-to-pay per specialist builder and the curation-transfer decay rate — to convert a credible structural tailwind into a quantified market.

INVESTMENT THESIS

Deployment economics are fragmenting AI from one generalist into a near-doubling population of small specialists (2M+ open models in 2025, median still ~0.41B params), and the 2026 frontier has made capability-per-active-parameter the design default — GLM-4.5 sits on the SWE-bench-vs-parameters Pareto frontier with far fewer parameters than competitors [16] and MiniMax-M2 activates only ~4.3% of its parameters per token [17]. Because the optimal corpus is compute- and precision-relative (and distillation scaling laws make each deployment regime its own teacher/student data job [18]), each specialist is a separate curation-plus-eval customer — fragmentation multiplies data JOBS, not just models. The right bet is a productized curation+eval engine over proprietary, verifiable vertical corpora (clinical claims/coding and similar), where every edge/cloud/precision variant is a new high-margin job on the same owned data. The hard caveat, which underwrites the diligence question: the literature proves a demand MULTIPLIER (more distinct buyers/jobs), not willingness-to-pay. Fund the data-and-eval moat, but gate the round on real per-builder pricing and curation-transfer decay — a 10x customer count at near-zero ARPU is not a business.

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